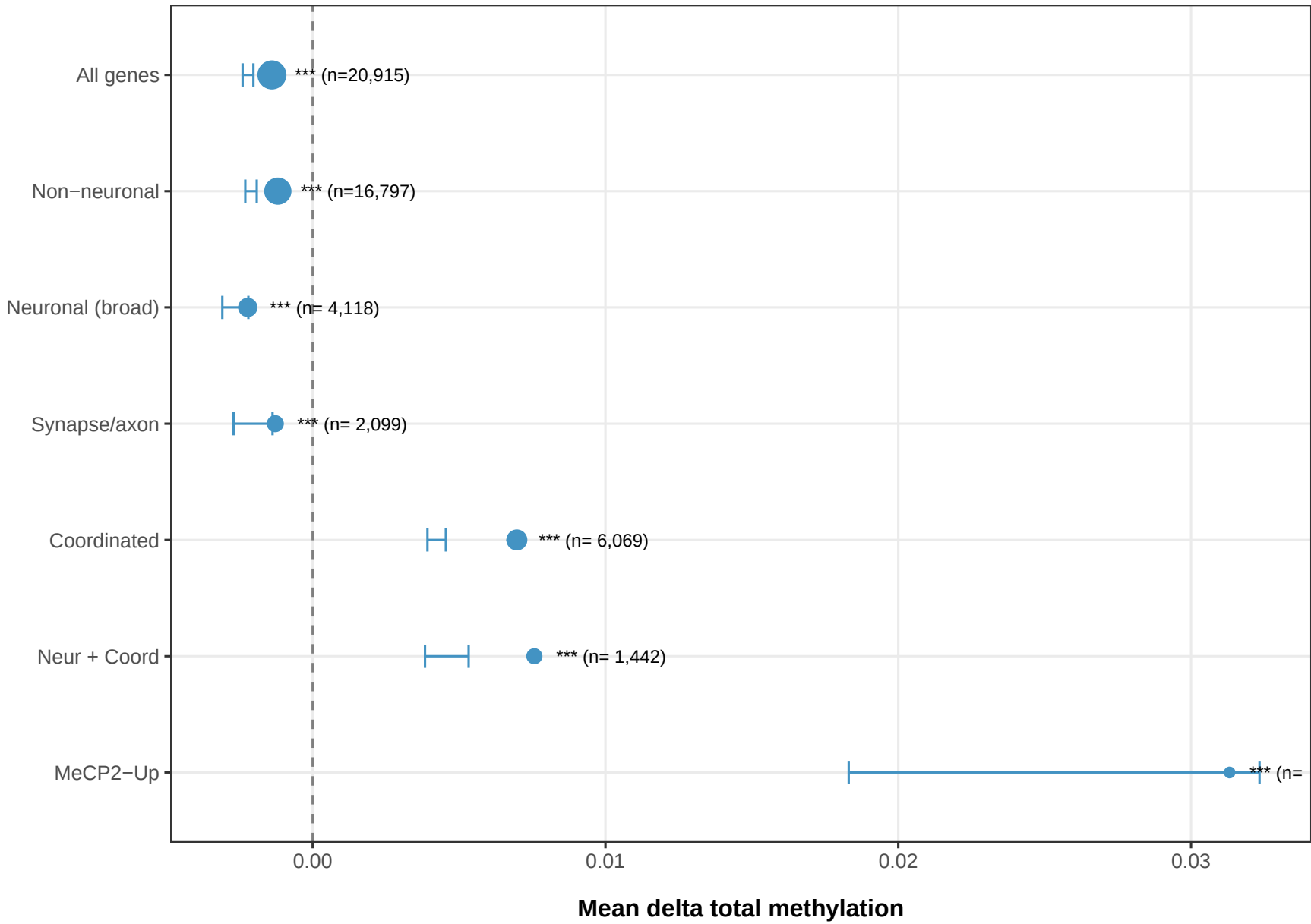


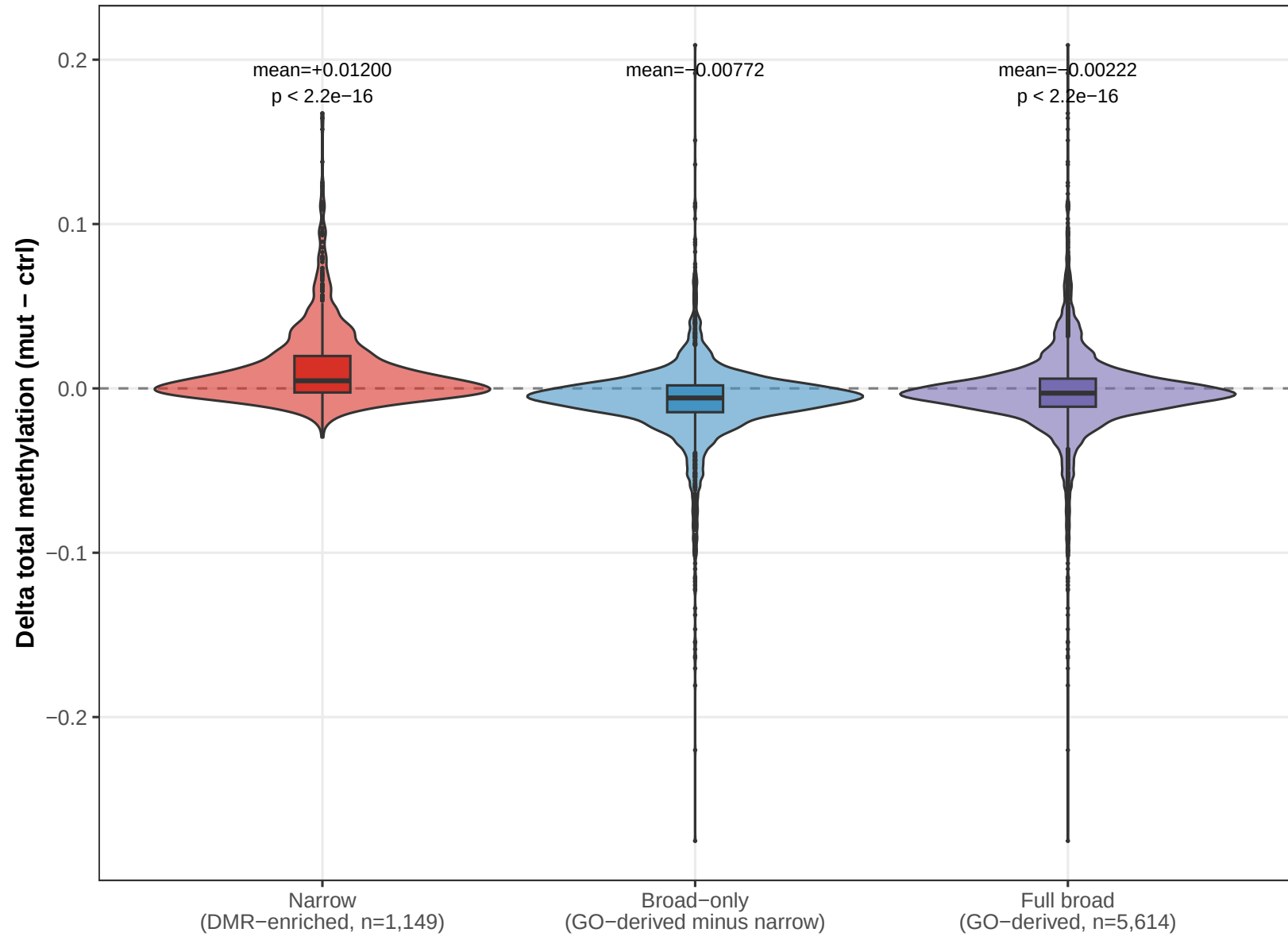
Section 78: Stoichiometry & Mechanism with Unbiased Neuronal Set

Broad neuronal: 4,118 genes (GO-derived) | Narrow: 1149 (DMR-enriched) | Synapse/axon: 2,099

Total methylation (mC+hmC) change in BAP1-KO
Mean delta (mut – ctrl) with Wilcoxon 95% CI | dashed = conservation

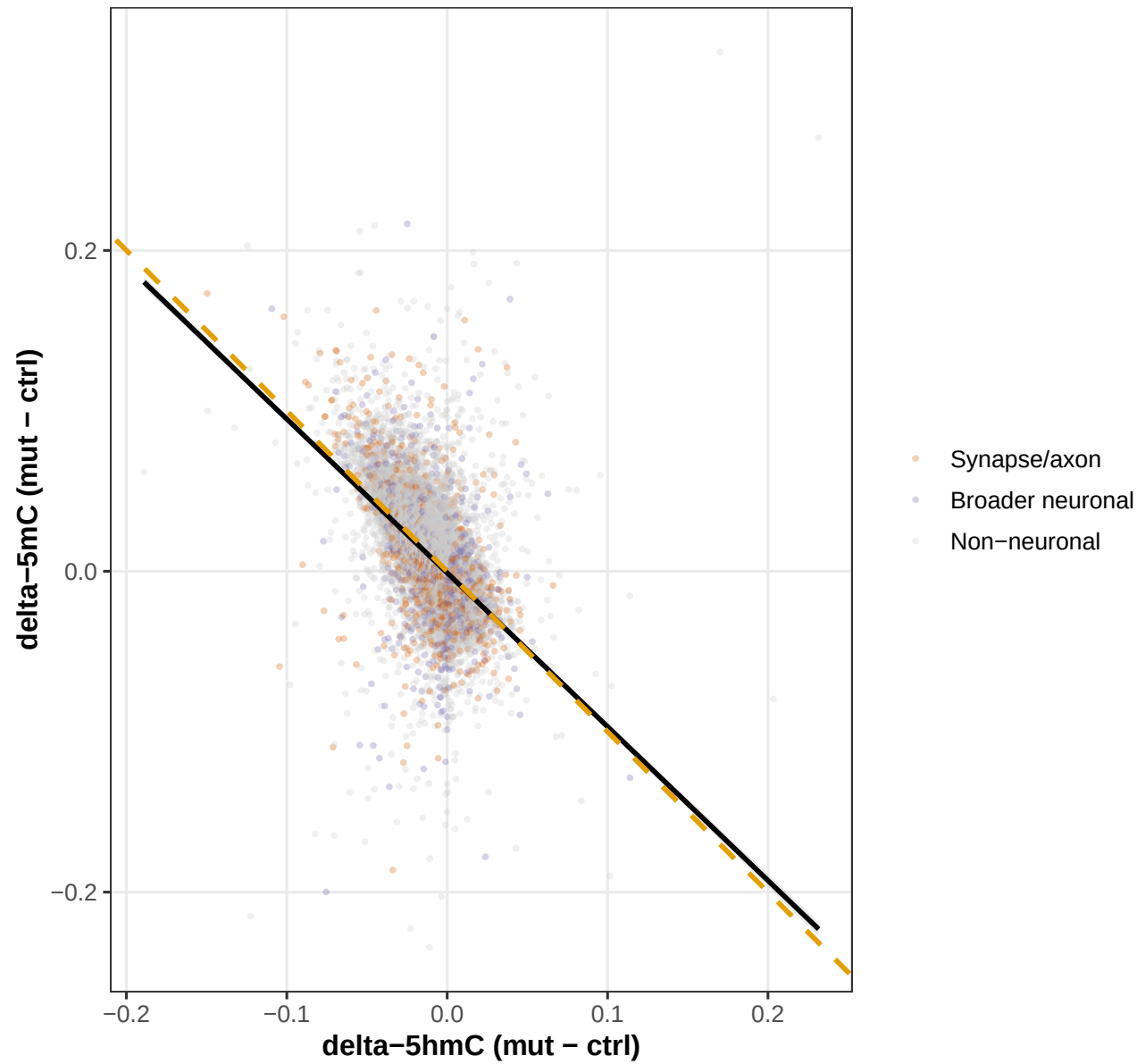


Bias demonstration: narrow vs broad neuronal gene set
Narrow set was derived from DMR enrichment (circular selection bias)
Broad set is GO-derived from org.Mm.eg.db (unbiased by methylation data)



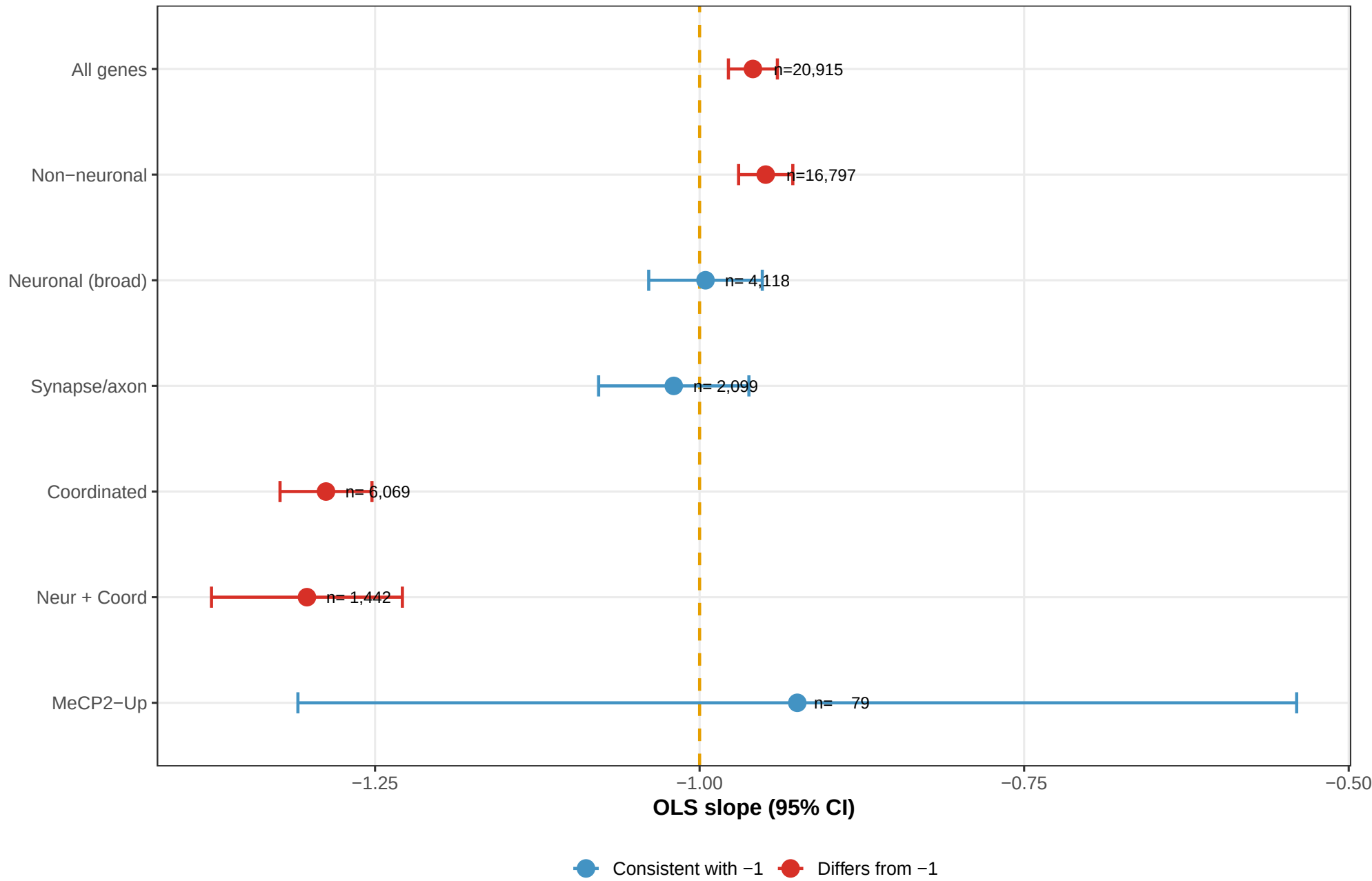
Stoichiometry: delta-5mC vs delta-5hmC

Black = OLS (slope=-0.959, R2=0.3218) | Orange = reference -1 (dehydroxymethylase)



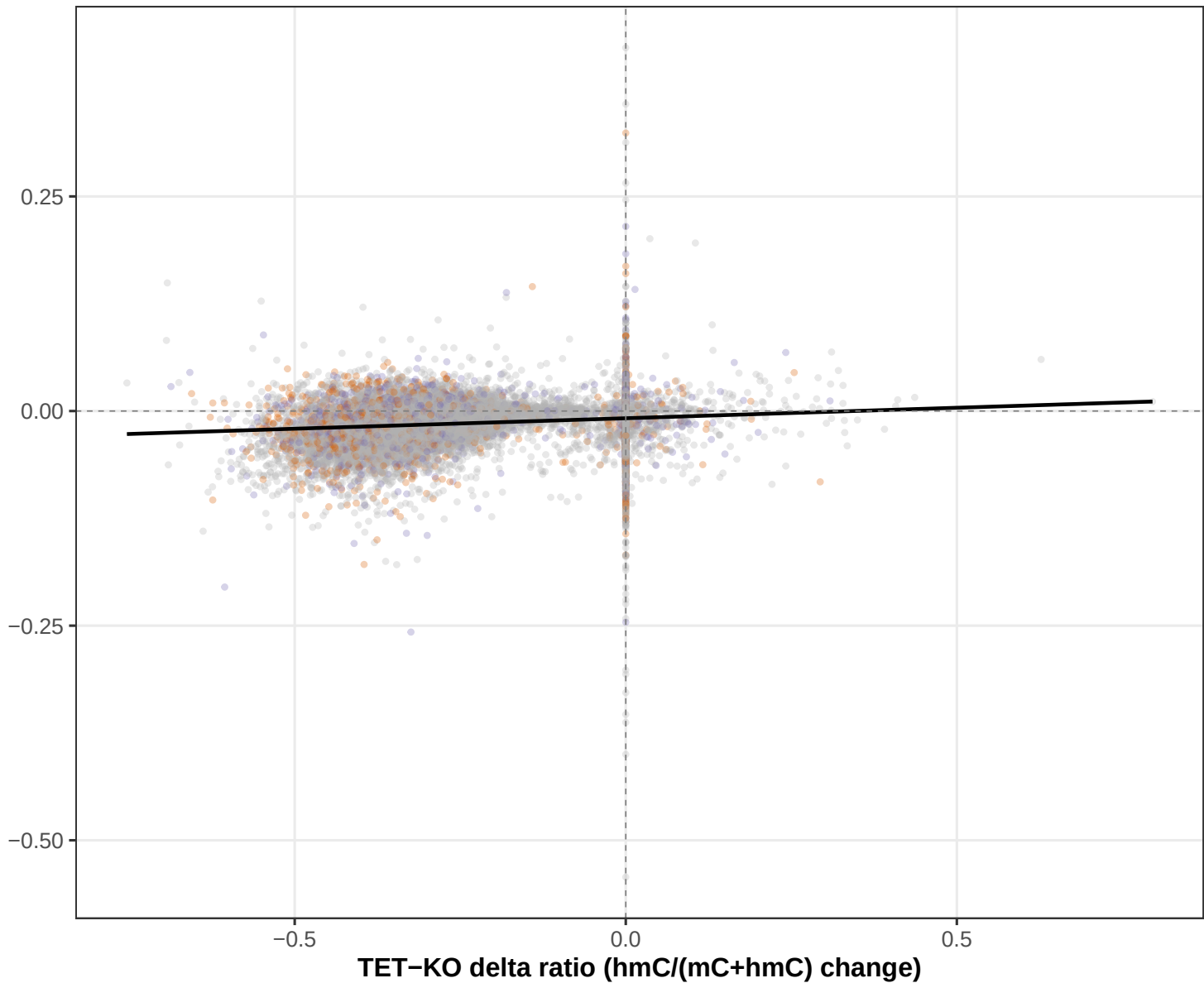
Stoichiometry slopes: delta-5mC ~ delta-5hmC

Orange dashed = -1 (stoichiometric dehydroxymethylase) | Red = significantly differs

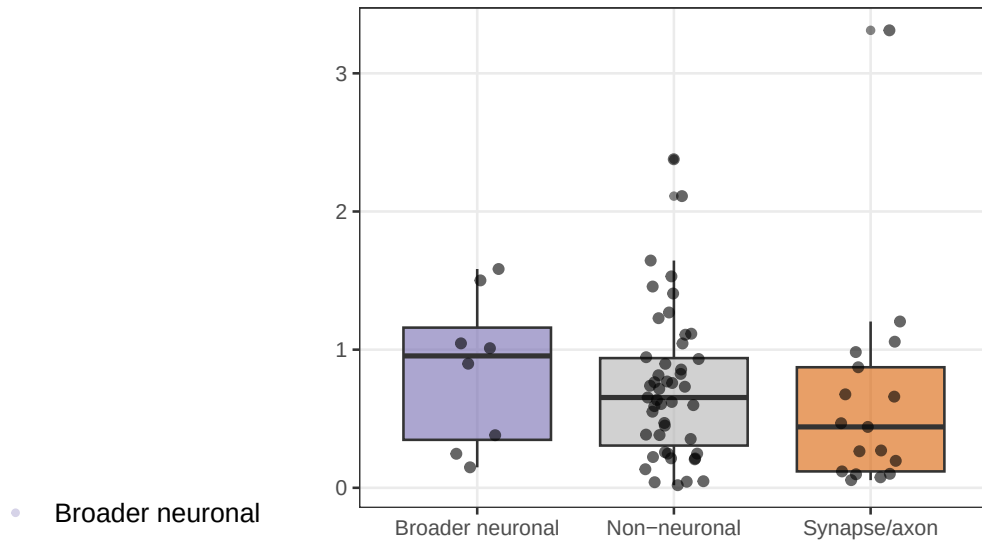


BAP1-KO vs TET-KO: TET efficiency change

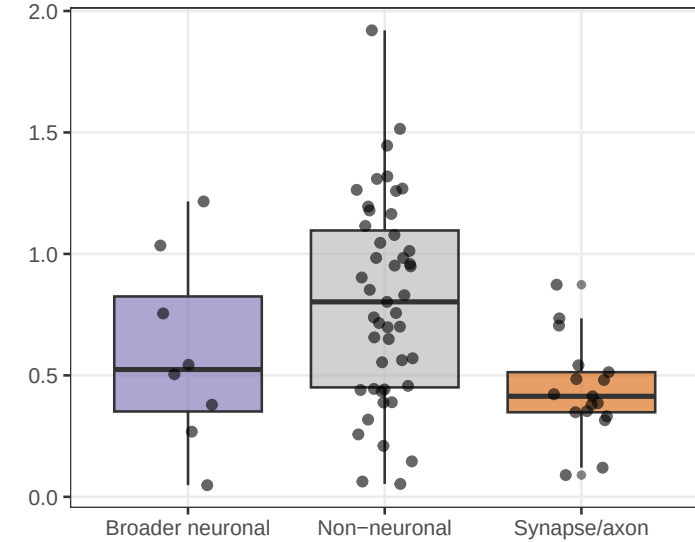
Spearman rho=0.220, p < 2.2e-16, n=20,851



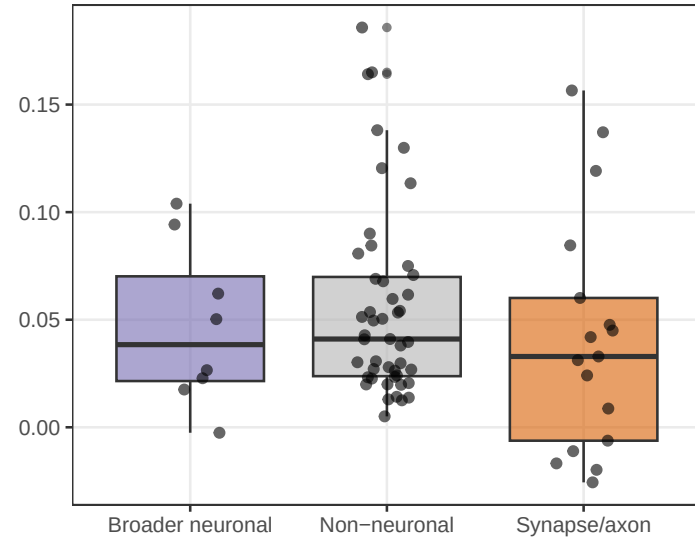
MeCP2 fold change



K119ub log2FC



5mC difference



5hmC difference

